

A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

OSTEOMYELITIS & SKIN AND SOFT TISSUE INFECTION (SSTI)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

CRP	C-Reactive Protein
ESR	Erythrocyte Sedimentation Rate
FQ	Fluoroquinolone
MRSA	Methicillin-resistant Staphylococcus aureus
MSSA	Methicillin-susceptible Staphylococcus aureus
OS	Osteomyelitis
Spp	Species
PEP	Post-Exposure Prophylaxis
SMX/TMP	Sulfamethoxazole/Trimethoprim
WBC	White Blood Cells

Licensed to Shreyansh Salunke (ID: 4477)

PART I: SKIN AND SOFT TISSUE INFECTIONS (SSTI)

Objectives

- 1 Outline etiology and causative pathogens of SSTI
- 2 Define clinical signs and symptoms of SSTI
- 3 Describe management of skin and soft tissue infections including MRSA skin infections

CASE 1

Tumor, Rubor, Dolor, Calor...

LS, a 67-year-old female, presents to the emergency department with a 3-day history of worsening left lower extremity redness, pain and swelling. She recalls an insect bite on the lower leg a few days prior to the onset of symptoms. On presentation to the hospital, her vital signs are stable and within normal limits. Her left calf is visibly more swollen than the right calf. She states her pain is 5-6 out of 10 (on a 10-point pain scale). Redness is noted and extends from just above the ankle to just below the knee. The area is tender to touch with non-pitting edema.

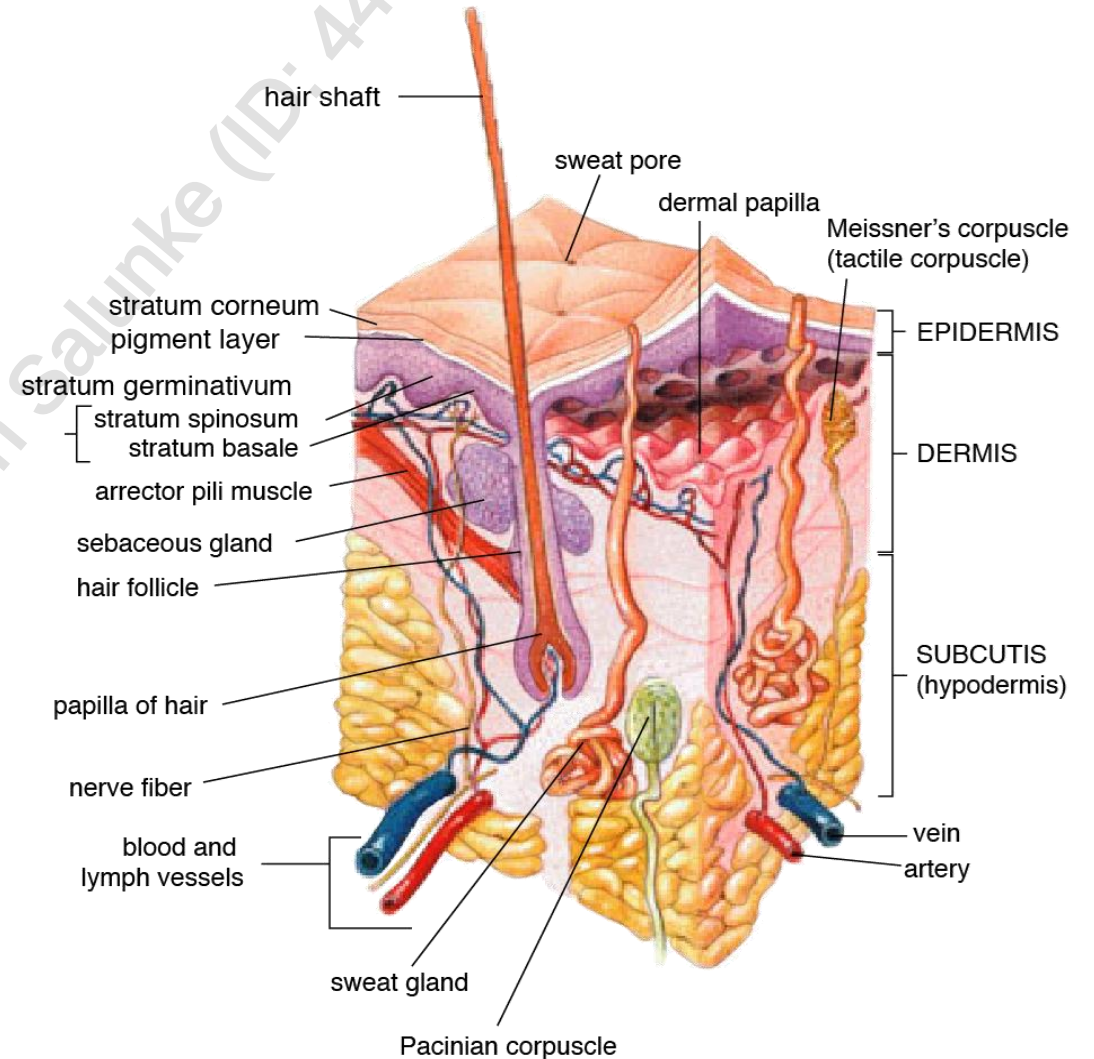
DEFINITIONS

Uncomplicated skin and soft tissue infections:

- Include: Folliculitis, impetigo, carbuncles, furuncles, cellulitis, small skin abscess

Complicated infections:

- Include: Large abscesses, ulcers, necrotizing fasciitis, bites and burns



PATHOPHYSIOLOGY

FYI

Competition of skin normal flora for growth: Causes infections when the protective and defense mechanisms of skin are deficient

Protective defenses of skin:

- **Protective surface of stratum corneum:** Acidic pH
- **Renewal of epidermis:** Disposal of contaminants
- **Sebaceous secretions:** Inhibition of reproduction of bacteria by free fatty acids
- **Normal skin flora:** Compete with foreign bacteria for growth
- **Microcirculation:** Provides nutrition and immune system response factors

Normal flora of skin:

- G⁺ bacteria: *Staphylococcus epidermidis*, *Staphylococcus aureus*, *Streptococcus spp*, *Corynebacterium spp*, *Propionibacterium spp*
- Yeast: *Candida spp*

Pathogens

Typical community-acquired SSTI pathogens

- *Staphylococcus aureus*
- *β-hemolytic streptococci* (groups A, B, C, G)

Less common SSTI pathogens

- *Enterococcus spp.*
- *Pseudomonas aeruginosa*
- *E. coli*, other *Enterobacteriaceae*
- Water, soil and zoonotic pathogens

RISK FACTORS

General risk factors contributing to SSTI include

- **Vascular insufficiency** (diabetes, peripheral vascular disease)
- **Indwelling devices** (catheters)
- **Immunocompromised patients** (transplants, cancer, AIDS, critically ill, elderly)
- **Epidermal skin break** (edema, trauma, surgery)
- **Socioeconomic status**
- **Obesity**
- **Prior skin and soft tissue infection**

Risk factors for CA-MRSA (*S. aureus*)

- **Opioid Use Disorder via injection, communal living environments, gym/athletics**

ASSESSMENT OF SEVERITY OF INFECTION

FEATURE	Mild/Uncomplicated	Severe/Complicated
Presentation	Slow progression	Acute/rapid progression
Tissues involved	Epidermis/dermis	Fascia, muscle, joint, bone
Systemic signs	Minimal	• Fever • Hypotension
Metabolic signs	Hyperglycemia	• Hyperglycemia • Acidosis • AKI

UNCOMPLICATED SSTI- NON-PURULENT VS PURULENT INFECTIONS

Non-purulent Infections

- Erysipelas
- Cellulitis

Purulent Infections

- Impetigo
- Folliculitis
- Purulent cellulitis +/- abscess

UNCOMPLICATED SSTI: NON-PURULENT INFECTIONS

PATHOGENS & CLINICAL SYMPTOMS

Erysipelas/Cellulitis

- **Erysipelas**: infection of the epidermal layer
- **Cellulitis**: Infection of the dermis, subdermal tissue

Pathogens

- **Normal host**: *Strep. spp.* (groups A,B,C,G) >>> *S. aureus* (including CA-MRSA),
- **Immunocompromised, diabetics**: Polymicrobial (G+, G-, anaerobes)

Clinical symptoms

- Hallmark signs: **Erythema, warmth, swelling, tenderness**
- Lymphadenopathy
- +/- abscess formation

UNCOMPLICATED SSTI: NON-PURULENT INFECTIONS

DIAGNOSIS AND CLASSIFICATION

Diagnosed by clinical examination and history

Cultures:

- superficial skin cultures are not helpful in
- blood cultures are rarely positive (<5%)

Classification:

Mild: local symptoms

Moderate: non-severe systemic symptoms

Severe: progression/failure of infection, severe systemic signs/symptoms, immunocompromised patients

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Her past medical history is significant for type II DM (HbA1c: 7.2%), hypertension, hypothyroidism and atopic dermatitis. She denies any travel, animal, soil or water exposure.

- What is your initial assessment of the patient's severity of illness and diagnosis?
- What is this happening to her?

UNCOMPLICATED SSTI: NON-PURULENT INFECTIONS

TREATMENT

Cellulitis/Erysipelas Treatment:

- **Non-pharmacological**: evaluation for skin abscess, drainage if present (see purulent infections)
- **Pharmacotherapy** (5 – 7 days, longer in severe/complicated):
 - **Mild**: *Penicillin*, Cephalexin, Cefadroxil, Clindamycin
 - **Moderate**: IV Penicillin, Cefazolin, Ceftriaxone, *Clindamycin*
 - **Severe**: [Piperacillin/tazobactam OR Cefepime] + Vancomycin

Considerations for treatment of CA-MRSA cellulitis

- **Treat empirically for CA-MRSA** if:
 - Non-response to anti-MSSA, anti-streptococcal coverage
 - Concern for severe/necrotizing infection

UNCOMPLICATED SSTI: FYI

SPECIAL CASES

Special cases of cellulitis:

Fresh water exposure:

- Cause: *Aeromonas spp.*, *P. aeruginosa*
- Treatment: Ciprofloxacin, Meropenem, Imipenem

Salt water exposure:

- Cause: *Vibrio vulnificus*
- Treatment: Doxycycline, Cefotaxime, Ciprofloxacin

Soil-contaminated wound:

- *P. aeruginosa.*, *Clostridium spp.* *Bacillus spp.*

Facial cellulitis:

- Cause: *Haemophilus Influenzae*, *S. pneumoniae*
- May be caused or complicated by otitis, sinusitis

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Her past medical history is significant for type II DM (HbA1c: 7.2%), hypertension, hypothyroidism and atopic dermatitis. She denies any travel, animal, soil or water exposure.

She has no known allergies

- What therapy would you suggest?

CASE 2

Honey...honey

MK, a 3-year-old female, presents to your family health team with 15-20 small, honey-crusted lesions on her left cheek/jaw. She complains of some itchiness and irritation but is otherwise in her normal state of health.

UNCOMPLICATED SSTI: PURULENT INFECTIONS

IMPETIGO

Impetigo

- Superficial, numerous nonfollicular pustules
- Occurs prominently on exposed areas of the face and extremities
- Common pathogens: *Staphylococcus aureus*, β -hemolytic *streptococcus species*

Risk factors

- Common in children, peak incidence: 2-5 years
- Equal in male and female
- Tropical regions, warmer months of the year
- Poor hygiene
- Minor skin trauma (insect bites)
- May be highly contagious, patients often colonized prior to infection

Clinical presentation

- Honey-crusted lesions, pustules rupture, then form thick crust

UNCOMPLICATED SSTI: PURULENT INFECTIONS

IMPETIGO

Bullous impetigo (impetigo subtype):

- Lesions with a central moist crust and outer zone of blister formation
- Common pathogen: *Staphylococcus aureus*
- Certain *S. aureus* strains produce a toxin that cleaves superficial skin layer

Differential diagnosis: Ecthyma

- Skin infection similar to impetigo, but more deeply invasive
- *Pathogens*: *S. aureus*, group A strep (GAS), other β -hemolytic streptococci implicated (groups C and G)

Treatment

- Mild: Topical: **Mupirocin or fucidic acid**
- Moderate/Severe: Oral: **cephalexin, cloxacillin, clindamycin**
- Consider CA-MRSA if rapidly progressing – if CA-MRSA, then use TMP-SMX, doxycycline, clindamycin

CASE 2

Honey...honey

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MK was prescribed mupirocin ointment to be applied twice a day for 5 days. Unfortunately, she developed additional lesions, some more than 1 cm in diameter, on her torso. She remains afebrile and systemically well.

She has a documented allergy to amoxicillin, which resulted in a maculopapular rash on day 5 after using it for an ear infection.

What therapy would you suggest?

UNCOMPLICATED SSTI: PURULENT INFECTIONS

FOLLICULITIS

Folliculitis

- Superficial skin infection localized at hair follicles

Pathogens

- Polymicrobial: *Strep*, *Staph*, *Candida spp.*, occasionally *Pseudomonas aeruginosa*

Risk factors

- Nasal carriage of *S. aureus*
- Pools, hot tubs (*Pseudomonas*)

Clinical symptoms

- Multiple clustered lesions ≤ 5 mm, erythematous, pruritic, pus

UNCOMPLICATED SSTI: PURULENT INFECTIONS

FOLLICULITIS

Treatment

- Lesions may spontaneously drain and resolve
- Topical treatment:
 - Warm compresses
 - Antibacterials (e.g. Mupirocin)
 - Antifungal agents
- Oral or parenteral antimicrobials is systemic infection (rare)

CASE 3

Bullseye...

SJ, a 26-year-old male, presents to his local pharmacy with a 4-day history of left arm pain and swelling. There appears to be a 2-cm fluctuant mass located on the anterior upper arm that is mobile and has surrounding redness of 5 cm in diameter.

He denies any fever, chills or other signs of systemic illness.

UNCOMPLICATED SSTI: PURULENT INFECTIONS

PURULENT CELLULITIS

Skin abscess

- Small to large purulent lesion containing pus, potentially associated with surrounding cellulitis

Furuncle:

- Small skin abscess extending through the dermis via hair shaft into the subcutaneous tissue
- Painful, tender and pus-filled. Often drains spontaneously.

Carbuncle:

- Coalescence of several inflamed furuncles
- May present with fever, commonly in patients with diabetes

UNCOMPLICATED SSTI: PURULENT INFECTIONS

PURULENT CELLULITIS

Pathogens

- Commonly *S. aureus*

Treatment:

- Mild: Incision and drainage, warm water or saline compresses for mild disease for 20-30 minutes QID. May consider antimicrobials for small skin abscesses to reduce recurrence
- Moderate: incisions and drainage + TMP/SMX, doxycycline, clindamycin x 7-10 days
- Severe: incisions and drainage + vancomycin, linezolid, daptomycin (may also require polymicrobial coverage)

CASE 3

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He denies any fever, chills or other signs of systemic illness.

As a diligent community pharmacist, you refer him to the approach level of care for further management.

To whom would you refer?

What therapy would you recommend based on the type and severity of illness?

COMPLICATED SSTI

Definition and examples

- Involve **deep skin structures** (fascia, muscle layers)
- May require surgical intervention
- Involvement of perianal area
- Foot infection in diabetics
- Coexisting disease: Diabetes, immunocompromised, obesity

Types of CSSTI:

- **Necrotizing wounds:** Cellulitis, fasciitis
- **Bites, burns, and surgical wounds**
- **Ulcers:** Decubitus, diabetic ulcers

COMPLICATED SSTI

Animal bite wounds:

- Cat bites deeper than dogs, 40-50% become infected

Pathogens:

- *S. aureus*, *Streptococci*, mouth anaerobic pathogens (*Pasteurella spp.*, *Eikenella spp.*)

Treatment:

- 5-10 days: Amoxicillin-Clavulanate, Moxifloxacin + Clindamycin
- Patients may be considered for tetanus toxoid, rabies PEP

COMPLICATED SSTI

Human bite wounds:

- Deep wounds, commonly cause infection
- Vital to culture if deep wound

Pathogens:

- *S. aureus*, *S. epidermitis*, *Streptococci*, mouth anaerobes (*Peptostreptococci*), *S. viridans*, *Eikenella* spp.)

Empiric treatment:

- 5 days if early presentation
- Up to 14 days if later presentation/more complicated wound
- IV: Amoxicillin/clavulanate, Carbapenem
- Oral: Amoxicillin/Clavulanate, Doxycycline

COMPLICATED SSTI

FYI

Necrotizing SSTI:

- **Characterized by:**
 - Acute life-threatening infection
 - Fulminant, widespread necrosis of soft tissue
 - Systemic toxicity
 - High mortality

Examples:

- Necrotizing fasciitis/cellulitis
- Clostridial myonecrosis
- Fasciitis tends to infect deep into the fascia and muscle. Cellulitis is more superficial than fasciitis and is confined to the cutaneous and subcutaneous area of the skin

COMPLICATED SSTI

FYI

Pathogens causing necrotizing SSTI:

Polymicrobial:

- ~ 2/3 of the time
- G + and G-ve
- Anaerobes
- Surgical, decubitus ulcer, abscess sources
- Common in diabetics and vascular insufficiency

Monomicrobial:

- ~ 1/3 of the time
- Group A *Streptococcus*: “flesh eating bacteria”
- *S. aureus*, CA-MRSA
- *Clostridium perfringens* (gas gangrene)

COMPLICATED SSTI

FYI

Etiology of necrotizing SSTI:

- Trauma, surgical site infection, perianal abscess, decubitus ulcer

Risk factors:

- PWID, immunosuppression, diabetes mellitus, obesity, elderly, alcohol, peripheral vascular disease, AIDS, cancer

Treatment:

- Aggressive debridement, supportive care and IV antibiotic therapy
- Empiric therapy:
 - Broad spectrum: Carbapenems, Piperacillin/Tazobactam
 - MRSA: Vancomycin, Daptomycin, Linezolid
 - Inhibit toxin production: Clindamycin (added on to above therapy)

COMPLICATED SSTI

FYI

Burn wounds:

Pathogens:

- Early organisms: Skin flora (*Staph/Strep spp.*)
- Nosocomial infections:
 - Gram negative (*Pseudomonas spp.*)
 - Resistant pathogens (*SPACE organisms*)

Considerations in severe burns:

- Resistance patterns
- Patient metabolism of drugs may be increased

COMPLICATED SSTI

FYI

Decubitus ulcers:

- Pressure sores
- Tissue ischemia and neural damage can lead to necrosis
- Not all ulcers are infected ulcers

Classification (rated by clinical signs):

- 1st degree: Erythema
- 2nd degree: Induration, blisters
- 3rd degree: Shallow ulcers
- 4th degree: Deep necrosis of fat and muscle
- 5th degree: Bone destruction

Pathogens:

- Polymicrobial: *S. aureus*; G⁻ infections: *Enterobacter*; Anaerobes: *Bacteroides spp.*

COMPLICATED SSTI

FYI

Decubitus ulcer etiology: Pressure, friction, moisture

Risk Factors:

- Paralysis
- Immobilization
- Malnutrition
- Elderly

Treatment:

- Remove pressure point, debridement
- **IV antimicrobial therapy** if signs of cellulitis or infection:
 - Amoxicillin/clavulinate
 - Piperacillin/Tazobactam, carbapenems
 - 3rd generation cephalosporin or fluoroquinolone +/- metronidazole
 - Consider risk factors for MRSA: **Vancomycin**

Complications: Cellulitis, osteomyelitis, sepsis

TRAUMA WOUNDS/COMPOUND FRACTURES

- Classified as **contaminated and/or dirty wounds**
- In these type of wounds, **antimicrobials** are considered **therapeutic not prophylactic**

Recommended empiric antibiotics:

- **Depend on trauma location and extent**
- Visceral and mucosal sites: Require broad-spectrum agents (cover for Pseudomonas if water related wound, metronidazole for soil contaminated wounds)
 - **Ciprofloxacin + metronidazole**
 - **Carbapenem**
 - **Cefazolin + aminoglycoside**
 - **Piperacillin-tazobactam**

PART II: OSTEOMYELITIS

Objectives

- 1 Outline etiologies of osteomyelitis (OS) with respect to usual age, site of infection, risk factors, common organisms and clinical findings
- 2 Understand clinical signs and symptoms and diagnosis of OS
- 3 Discuss antibiotics that can be used for OS treatment including efficacy, side effects, dosing regimen
- 4 Develop a therapeutic plan and monitoring plan for treating OS

CASE 4

That's the rub...

NK, a 72-year-old female, was transferred from a long-term care facility for intermittent fevers and increasing discharge from a known pressure injury on her left ankle. Upon investigation, she has a mildly elevated white blood cell count (13.5×10^9 cells/L), with a C-reactive protein level of 54 mg/L. Upon further evaluation, an MRI reveals an increasing signal in her ankle, indicative of osteomyelitis.

Her past medical history is significant for chronic kidney disease (baseline SCr is 142 mmol/L), depression, mild cognitive impairment and rheumatoid arthritis.

She has no known allergies.

DEFINITIONS

- Osteomyelitis is the inflammation of bone marrow and surrounding bone caused by infection
- Different types of osteomyelitis:
 - Hematogenous infection (...from the blood)
 - Contiguous infection (...from another site of infection)
 - Vascular insufficiency (...from tissue open to the environment)
 - Chronic osteomyelitis (...long lasting osteomyelitis)
 - Prosthetic joint infection (...bone infection related to hardware)

COMMON BACTERIAL CAUSES OF OSTEOMYELITIS

HEMATOGENOUS	CONTIGUOUS (ADJACENT SITE)	VASCULAR INSUFFICIENCY
Typically monomicrobial • <i>S. aureus</i> • Streptococci • People who inject drugs: <i>Pseudomonas</i> • Sickle cell anemia usually, <i>Salmonella</i> • Tuberculosis	<i>Mono- or polymicrobial</i> • <i>S. aureus</i> • G⁻ bacilli: <i>Pseudomonas</i>, <i>Proteus</i>, <i>Klebsiella</i>, <i>E. coli</i> • Anaerobic: <i>B. fragilis</i>	Typically polymicrobial • <i>S. aureus</i> • Beta-hemolytic streptococci • Coagulase-negative staphylococci, • G⁻ and anaerobic organisms

CLINICAL PRESENTATION

FEATURE	HEMATOGENOUS	CONTIGUOUS (ADJACENT SITE)	VASCULAR INSUFFICIENCY
Age of onset	< 20 years: (most often in children), > 50 years, vascular access	> 40 years (more in adults)	> 40 years
Sites of infection	Long bones, vertebrae	Femur, tibia, skull, mandible	Feet

CLINICAL PRESENTATION

Dependent on type of osteomyelitis

- Fever
- Pain/inflammation (less common with vascular insufficiency)
- Discharge
- Joint/bone instability
- Sinus tract formation

CLINICAL PRESENTATION

FYI

Imaging:

- Radiographs are often normal in the first two weeks, therefore MRI is preferred to detect early infections
- May take a long time for systemic symptoms to develop
- Erythrocyte sedimentation rate (ESR):
 - Elevated ($> 20\text{mm/h}$)
 - ESR: Rate at which erythrocytes settle to the bottom of a test tube because of gravity
 - Response to fibrinogen levels in the blood
 - Increased abnormally in acute or chronic inflammation, infections, tissue necrosis, etc.
 - Non-specific
- Serum C-Reactive Protein (CRP):
 - A measure of systemic inflammation and should be elevated (more sensitive than ESR)
 - Synthesized in the liver in response to inflammatory process ($> 2.0\text{ mg/dL}$)

CLINICAL PRESENTATION

FYI

Complete blood count (CBC)

WBC count: slightly increased, i.e. leukocytosis (more drastic in acute infections)

Culture

Bone aspiration/debridement

- If abscess located, the fluid is drained and cultured, and gram stain performed

Blood cultures

RISK FACTORS

HEMATOGENOUS	CONTIGUOUS (ADJACENT SITE)	VASCULAR INSUFFICIENCY
Bacteremia: The bacteria can travel via blood to certain bone areas, colonizing and causing infection	• Surgery • Trauma (penetrating injury/open fractures) • Cellulitis/Abscess	• Diabetes • Peripheral vascular disease (PVD)

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Her past medical history is significant for chronic kidney disease (baseline SCr is 142 mmol/L), depression, mild cognitive impairment and rheumatoid arthritis.

She has no known allergies.

What type of osteomyelitis does this represent?

What are the likely pathogens?

GOALS OF THERAPY

- ✓ Eliminate undesirable signs and symptoms
- ✓ Prevent complications of disease and therapy
- ✓ Reduce recurrence risk of the infection

NON-PHARMACOLOGICAL MEASURES

- Surgical decompression and debridement
- Hyperbaric oxygen (HBO)
- Wound care
- Pressure offloading
- Revascularization

Critical to maximizing success of therapy!
(Particularly in vascular insufficiency osteomyelitis)

GENERAL PRINCIPLES OF PHARMACOLOGICAL THERAPY

Duration of therapy:

- **4 to 8 weeks of antibiotics** (longer in infections with retained hardware or in chronic infection)
- *Switch to oral antibiotics as soon as feasible if patients able to swallow and are clinically stable (i.e. not hypotensive) Best studied: fluoroquinolones, clindamycin, TMP/SMX. Emerging evidence for doxycycline and beta-lactams.

Other considerations:

- Consider **pain management** (e.g., acetaminophen, NSAIDS, opioids)

PHARMACOLOGIC MEASURES

Treatment of hematogenous and contiguous osteomyelitis:

- Targeted initial treatment:
 - *Staphylococcus aureus* (MSSA): cefazolin, cloxacillin
 - *Staphylococcus aureus* (MRSA): vancomycin, daptomycin, linezolid
 - *P. aeruginosa*: ceftazidime, piperacillin/tazobactam, cefepime, meropenem, imipenem/cilastatin
- **IV therapy x 7-10 days (or sooner).** If patient responds, consider step down to oral x 6 weeks (OVIVA trial),
- Duration: **at least 4-6 weeks** (oral or IV) or until ESR/CRP has normalized.
- Shorter-course therapy in children (20 days) possible

PHARMACOLOGIC MEASURES

Treatment of vascular insufficiency osteomyelitis:

- Empiric treatment (examples dependent upon prior antimicrobial exposure, resistance)
 - *Ceftriaxone/Cefotaxime +/- vancomycin +/- metronidazole*
 - *Amoxicillin/clavulanate (IV) +/- vancomycin*
 - *Piperacillin/tazobactam +/- vancomycin*
 - *Carbapenem +/- vancomycin*
- IV therapy or oral therapy for ≥ 6 weeks
- Example oral options include:
 - Cephalexin, cefadroxil
 - Clindamycin or Metronidazole + Levofloxacin
 - Moxifloxacin
 - Amoxicillin/clavulanate
 - TMP/SMX +/- clindamycin or metronidazole
 - Doxycycline + FQ
 - Linezolid + FQ

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Her past medical history is significant for chronic kidney disease (baseline SCr is 142 mmol/L), depression, mild cognitive impairment and rheumatoid arthritis.

She has no known allergies.

What treatment advice should you provide?

PHARMACOLOGIC MEASURES

Treatment of prosthetic joint infection (note: not covered in detail)

Commonly affected joints

- Knee, hip, shoulder, elbow, etc.
- Devices made up of metallic alloys, cemented to bone

Etiology: *Staph* (most common), followed by *Strep*, G –ve

Signs and symptoms: Joint loosening, sinus tract formation, swelling, erythema and tenderness over the prosthetic joint, neutrophils in the joint fluid

Antibiotic treatment: **Must cover *Staph* and *Strep*** but targeted towards microbiology. Addition of rifampin possible if Staphylococci and hardware remains in place

Surgery: May be needed to remove the prosthetic joint and give antibiotics for 6 weeks to treat the prosthetic joint infection. If no surgery, or if joint replaced into infected material, at least 12 weeks

MONITORING PARAMETERS

- Ensure patient **responding to IV antibiotic** (WBC & fever ↓, less drainage, reduced CRP)
- **Monitor Cr, trough levels every 7days if using vancomycin** to avoid nephrotoxicity
- Monitor hematologic, dermatologic and other complications of long-term antimicrobials

THERAPEUTIC TIPS

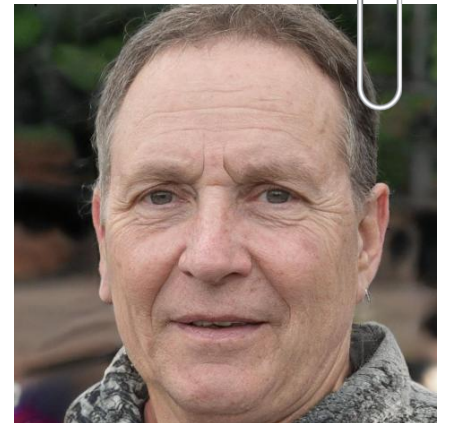
- Start IV antibiotic empirically only if indicated and based on suspected organisms and adjust therapy once cultures are available
- Duration of antibiotic is typically **4-8 weeks**. Longer therapy may be warranted if no improvement
- Consider **change to PO** antibiotic once WBC normalized, decreased pain, no complications seen (i.e. necrosis) only if antibiotic has good F. Increasing evidence for oral beta-lactams with majority of literature supporting non-beta-lactam therapy.
- Osteomyelitis therapy, especially in vascular insufficiency, **requires multiple types of interventions, antimicrobials alone will not solve all issues**

CASE 5

Mr. L is a 67-year-old man with a 20-year history of type 2 diabetes mellitus complicated by peripheral neuropathy and peripheral arterial disease. He presents to hospital with a 3-week history of a non-healing ulcer over the plantar surface of his right foot beneath the first metatarsal head. The ulcer measures 3 cm in diameter, probes to bone, and has surrounding erythema and mild purulent drainage. He is afebrile. Laboratory results show WBC $11.8 \times 10^9/\text{L}$, CRP 92 mg/L, HbA1c 6.9%, and serum creatinine 110 micromol/L (eGFR ~60 mL/min). Plain radiograph demonstrates cortical erosion of the first metatarsal consistent with osteomyelitis. Wound cultures are pending. He has no history of MRSA colonization. Current medications include metformin, ramipril, atorvastatin, ASA 81 mg daily, and insulin glargine. He has no drug allergies.

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Mr. L is a 67-year-old man with a 20-year history of type 2 diabetes mellitus complicated by peripheral neuropathy and peripheral arterial disease. He presents to the hospital with a 3-week history of a non-healing ulcer over the plantar surface of his right foot beneath the first metatarsal head. The ulcer measures 3 cm in diameter, probes to bone, and has surrounding erythema and mild purulent drainage. He is afebrile. Laboratory results show WBC $11.8 \times 10^9/\text{L}$, CRP 92 mg/L, HbA1c 6.9%, and serum creatinine 110 micromol/L (eGFR ~60 mL/min). Plain radiograph demonstrates cortical erosion of the first metatarsal consistent with osteomyelitis. Wound cultures are pending. He has no history of MRSA colonization. Current medications include metformin, ramipril, atorvastatin, ASA 81 mg daily, and insulin glargine. He has no drug allergies.



CASE 5 - QUESTIONS

1. Which of the following is the MOST appropriate empiric antimicrobial regimen for this patient's vascular insufficiency osteomyelitis while awaiting culture results?
 - a. cephalexin
 - b. vancomycin
 - c. piperacillin–tazobactam
 - d. doxycycline

2. Bone culture results return 5 days later revealing methicillin-susceptible *Staphylococcus aureus* (MSSA) only. The patient is clinically stable following debridement. Which targeted regimen is MOST appropriate?
 - a. cefazolin
 - b. ciprofloxacin
 - c. vancomycin
 - d. trimethoprim–sulfamethoxazole

CASE 5 - QUESTIONS

3. In addition to the antimicrobial therapy, which of the following is the MOST important additional therapeutic addition to care of this patient?
 - a. Incision and drainage
 - b. Pressure offloading
 - c. Control of diabetes
 - d. Below knee amputation

4. What is the MOST appropriate duration for therapy of the vascular insufficiency osteomyelitis in this patient?
 - a. 2 weeks
 - b. 3 weeks
 - c. 6 weeks
 - d. 12 weeks

CASE 6

JC is a 19-year-old female who came into the ER with a painful, inflamed right arm. JC noticed some pain and irritation a week ago after she scratched her arm after falling while playing indoor soccer. The pain has worsened over the past few days, and she went to the student medical center, where her wound was cleaned and she was prescribed cephalexin 500 mg PO Q6H x 7 days. JC presents to the ER with new purulent discharge after 3 days of cephalexin therapy and increasing tenderness/warmth and erythema. JC previously had an appendectomy in 2015. She has no known allergies, and she is up-to-date on her immunizations.

Objective findings:

- BP 129/70 mmHg, T 37.4°C
- Right arm red, erythematous and tender to touch; some pus oozing from the wound
- Notable fluctuant area (approx. 2cm)



CASE 6 - QUESTIONS

1. JC is most likely experiencing:
 - a) Cellulitis and abscess
 - b) Cellulitis
 - c) Necrotizing Fasciitis
 - d) Osteomyelitis
2. What is a reasonable treatment plan for JC?
 - a) Continue cephalexin
 - b) Switch to ciprofloxacin
 - c) Incision and drainage alone
 - d) Incision and drainage and doxycycline
3. For how long should JC be treated with antimicrobial therapy?
 - a) 3 days
 - b) 5 days
 - c) 7 days
 - d) 10 days

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CHANGE LOG

June 2025

- Slide 50: Minor edits, and updated treatment duration and antibiotic options to match Diabetes Canada foot care chapter
- Slide 61: Updated references

July 2025

- Significant edits made throughout
- Legend added and reference updated
- Case study questions throughout

August 2025

- Slide 41: Changed 'IV drug user' to 'people who inject drugs'

October 2025

- No content changes made

March 2026

- Slide 2: Updated NAPRA competencies
- New references added
- Clarifications for PO beta-lactams added to osteomyelitis
- Minor text revisions for clarity
- Slide 51 deleted
- Case 5 changed

April 2026

- Slide 20: Removed penicillin from treatment options